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Forward Deployed Engineer · Conversational AI, RAG and Multi-Agent Systems

SUMMARY

Forward Deployed Engineer who starts with the customer problem, not the model. Twelve years running B2B apparel distribution taught me to read buyer demand, scope requirements with stakeholders and own outcomes end to end; since 2024 I have applied the same discipline to conversational AI, RAG and multi-agent systems. Nine systems built end-to-end, six live in production, each backed by its own evaluation suite rather than assumed results. Example: cart abandonment is often a wish to bargain, so I built a Hinglish agent that negotiates like a shopkeeper.

SKILLS

- **Languages:** Python, SQL
- **Agent & LLM frameworks:** LangGraph (supervisor-router, interrupt() escalation, checkpointing), CrewAI, PageIndex (hierarchical retrieval), PipeCat (prototyping)
- **Models:** Claude (Sonnet, Haiku, Bedrock), GPT-4o, BERT / DistilBERT (fine-tuned classification, emotion, re-ranking), MuRIL (multilingual intent), Qwen 0.5B (fine-tuned), Jev (hosted classifier)
- **Retrieval & storage:** ChromaDB, BERT re-ranking, PageIndex, SQLite, semantic caching
- **Evaluation & safety:** DeepEval, golden test sets, LLM-as-judge, macro-F1 and latency benchmarking, pass@k reporting, label-quality audits, prompt-injection testing, output guardrails
- **Cloud & deployment:** AWS (ECS Fargate, EC2, S3, CloudFront, Bedrock), Azure, Railway, Vercel, HuggingFace Spaces
- **Integrations & APIs:** Twilio, Deepgram, ElevenLabs, Whisper, Sarvam AI, Shopify API and webhooks, Apify, UN Comtrade API, REST (GET/POST, CSRF-token flows)
- **Analytics:** Power BI, Tableau, Shopify Analytics, Meta Ads Manager

PROJECTS

Multi-Agent Customer Support Platform · LangGraph, DistilBERT, Jev, AWS

- Built a LangGraph supervisor-router that classifies support tickets and delegates to department agents, with interrupt()-based human escalation, SQLite checkpointing and semantic caching; deployed on AWS (ECS Fargate, EC2, S3, CloudFront, Bedrock).
- Benchmarked a fine-tuned DistilBERT router against Jev: DistilBERT led on macro-F1 (0.89 vs 0.56) and median latency (15 ms vs 354 ms), but LLM-judge review rated Jev's routing defensible 93% of the time vs 73%.
- Audited the training data and found only 71% of ground-truth labels held up; switched production routing to Jev and caught a style shortcut with handwritten counter-example tickets.

Dependency Rescue · LangGraph, sandboxed execution, DepBench

- Built a LangGraph agent that repairs broken dependency upgrades: reproduces the failure in an isolated sandbox, proposes an atomic fix, rejects unsafe or duplicate edits, reruns the test suite and stops at a human-approval gate with a reviewable diff.
- Evaluated on real breaking upgrades from DepBench, 3 runs per case: 2 of 5 dev cases fixed (pass@3), 3 of 15 full-fix runs at ~\$0.05 per run; held-out evaluation in progress.
- Eval harness exposed 3 bugs, all fixed: a mislabeled integrity failure, a substring match that flagged source files as tests, and unreported run-to-run variance (now reported as pass@3).

Legal RAG Chatbot · PageIndex, BERT re-ranker, GPT-4o, DeepEval

- Built an assistant for Indian legal acts (BNS, BNSS, BSA, DPDP Act 2023): an Act router, then PageIndex hierarchical retrieval, BERT re-ranking and GPT-4o generation.
- Built a 37-case DeepEval golden suite with a Bedrock Claude judge: Faithfulness 0.97 and Answer Relevancy 0.98 (36/37 pass each), plus retriever, safety, latency, cost and reliability evals.

BargainAI — Hinglish Negotiation Chatbot · MuRIL, PageIndex, Shopify

- Replaced generic abandoned-cart discounts with shopkeeper-style Hinglish bargaining; a MuRIL intent classifier trained on 51,000 Flipkart reviews detects hesitation type (price, quality, comparison, fit) before PageIndex retrieval over 10 sales and negotiation books.
- Hardened for production with a fixed-price placeholder and regex guardrail so the LLM cannot invent prices, Shopify webhook deduplication, two-stage close detection and single-use 24-hour discount codes.

RecallAI — Wellness Voice Agent · Twilio, Deepgram, BERT, ChromaDB, Claude, ElevenLabs

- Built a phone-based wellness companion: Twilio → Deepgram → BERT emotion detection → ChromaDB retrieval over therapy books → Claude → ElevenLabs playback, with Hindi support via Sarvam AI.
- Traced distorted audio to ElevenLabs ignoring the requested output format and fixed it; measured ~1-2 s sequential-API latency per turn and am prototyping a PipeCat streaming pipeline.

NiryatAI — Indian Export Intelligence Platform · Multimodal RAG, Whisper, UN Comtrade, Apify

- Built multimodal RAG over 114 files (~20,000 chunks) that answers exporters' questions on HS codes, schemes, costing, documentation and buyer leads, replacing 5+ government and trade portals; Whisper handles audio queries.
- Fixed data-integrity issues in the UN Comtrade pipeline with customs and transport-mode filters; reverse-engineered DGFT TradeStat (no official API) via CSRF-token GET and form POST; added Apify buyer leads across 41 countries behind an email-capture gate.

AI Complaint Intelligence Agent · CrewAI, 7 fine-tuned BERT classifiers, DeepEval

- Built four CrewAI agents (classifier, priority ranker, trend detector, report generator) over 7 fine-tuned BERT classifiers trained on 51K reviews, turning raw customer reviews into prioritized business intelligence.
- An 8-part DeepEval readiness suite found 3 release blockers, all fixed: truncated reports from an unset token limit, a prompt-injection surface and model reloads on every call. Migrating to LangGraph with a typed state schema and CI-integrated evals.

HR People-Analytics Agent · Text-to-SQL, Claude Haiku, Qwen 0.5B

- Scoped requirements with an HR team and built a text-to-SQL agent for stakeholders' ad hoc questions, prototyped on a public 5,000-row dataset to keep employee data private.
- On 15 golden questions, Claude Haiku reached 73% execution accuracy vs 47% for a fine-tuned Qwen 0.5B.
- Across 12 prompt-injection tests, rewrote a flawed safety oracle and closed the one schema-exposure breach with a shared guardrail: zero breaches, no accuracy drop.

Multi-Agent Company Architecture · LangGraph

- Unifies the systems above as department agents (Finance, HR, Ticketing, Export, Legal) under one orchestrator, with a verifier that checks grounding before responding or escalating to a human.

WORK EXPERIENCE

Independent AI Engineering (iMarketlyze, sole proprietorship) · Pune · 09/2024 – Present

- Vehicle for the client and portfolio work listed under Projects; scoped the HR People-Analytics Agent directly with a client's HR team.

Teespirit — Performance Marketing & Growth Analyst · Pune · 01/2022 – 03/2024

- Ran 140+ Meta ad campaigns on ₹29.5L of spend, driving ~3,966 orders, 24M+ reach, 47M+ impressions and 160K+ link clicks.
- Built Shopify Analytics and Power BI dashboards tracking CPC, CPM and add-to-cart funnels daily; paused underperformers within 48 hours and reallocated spend to top-quartile creative.
- Owned the full creative-to-conversion funnel: ad creative, A/B-tested visuals, audience targeting, landing-page optimization and budget allocation.

Kishor & Company — Business Head, B2B Apparel Wholesale · Maharashtra / Karnataka / Andhra Pradesh · 01/2012 – 01/2022

- Grew wholesale distribution from 40 to 200+ accounts across 38 cities in 3 states, coordinating 4 field agents over ~2,000 km of territory.
- Retained 15 anchor accounts year after year by matching product recommendations to their customers' demand; closed deals across language barriers, including Telugu-speaking buyers.
- Built a Tableau dashboard on 3 years of SKU-level sales data; found plain shirts outperformed printed designs and shifted production to a high-margin plain-shirts line that became a regional category leader.
- Used seasonal demand analysis to pre-stock counter-intuitive winners (Royal Blue for May-June, Black for Dec-Feb), cutting peak-season stockouts and freeing working capital.
- Managed 125-order procurement cycles across 8 suppliers; flagged bottleneck suppliers early and pre-routed orders to protect on-time delivery.

Naukri.com — Senior Sales Executive · Pune · 01/2011 – 01/2012

- Onboarded an average of 10 employer clients per month onto the recruitment portal by prospecting companies and HR teams.

EDUCATION

- **MBA, Marketing** · Pune University, Pune · 06/2009 – 03/2011
- **BBA, Marketing** · Solapur University, Solapur

CERTIFICATIONS

- Agentic AI Training · Edureka · 01/2025
- Digital Product Management Specialization · Coursera · 01/2025
- Google Data Analytics Professional Certificate · Google · 01/2025